Uniqueness Representation Theorem

Theorem

Let $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ *be a basis for a vector space* V.

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Let $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ be a basis for a vector space V. Then, for every $\mathbf{v} \in V$, there is a unique set of $c_1, \ldots, c_n$ such that

$$\mathbf{v} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$$

Proof of Uniqueness Representation Theorem

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1. $\text{Span}\left(\mathbf{v}_1, \ldots, \mathbf{v}_n\right) = V$
2. $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are all linearly independent.

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Define $A = \begin{bmatrix} \mathbf{v}_1 & \ldots & \mathbf{v}_n \end{bmatrix}$, then since the columns of A span all of V, we know that we can solve $A\mathbf{c} = \mathbf{v}$ for every $\mathbf{v} \in V$.

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Define $A = \begin{bmatrix} \mathbf{v}_1 & \ldots & \mathbf{v}_n \end{bmatrix}$, then since the columns of A span all of V, we know that we can solve $A\mathbf{c} = \mathbf{v}$ for every $\mathbf{v} \in V$. Since the columns of A are linearly independent, we know $\mathbf{c}$ is unique. Putting this all together means

$$\mathbf{v} = A\mathbf{c} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n.$$

□

Basis as a Coordinate System

Definition

Let $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be a basis for a vector space V. Then, for every $\mathbf{x} \in V$, we define the $\mathcal{B}$ coordinates of $\mathbf{x}$ to be the unique scalars $c_1, \ldots, c_n$ such that $\mathbf{x} = c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n$.

Definition

We define the $\mathcal{B}$-coordinate vector of $\mathbf{x}$ to be:

$$[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix}$$

Coordinate System Example

Let $V = \mathbb{R}^{2\times 2}$ and consider the basis given by

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$$

And label them as follows

$$\mathbf{v}_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \mathbf{v}_4 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

Then

$$\left[\begin{bmatrix} 1 & 5 \\ -4 & 15 \end{bmatrix} \right]_{\mathcal{B}} = \begin{bmatrix} 1 \\ 5 \\ -4 \\ 15 \end{bmatrix}$$

Coordinate System Example pt. 2

Let $V = \mathbb{R}^3$, $F = \mathbb{R}$, and $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 3 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} \right\}$ be a basis of V.

Coordinate System Example pt. 2

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$$\left[\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right]_{\mathcal{B}}$$

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We want to solve the following augmented system:

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ -1 & 3 & -1 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right]$$

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$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ -1 & 3 & -1 & 0 \\ 1 & 1 & 0 & 1 \end{array} \right] \xrightarrow[R_3 = R_3 - R_1]{R_2 = R_2 + R_1} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 3 & 1 & 1 \\ 0 & 1 & -2 & 0 \end{array} \right]$$

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$$\xrightarrow{R_3 = \frac{1}{7} R_3} \left[\begin{array}{ccc|c} 1 & 0 & 2 & 1 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & 1 & \frac{1}{7} \end{array}\right]$$

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Coordinate System Practice

Let $V = \mathbb{R}^3$, $F = \mathbb{R}$, and $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} \right\}$ be a basis of V.

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Coordinate Mapping

Definition

Coordinate Mapping: The mapping $C : \mathbf{x} \mapsto [\mathbf{x}]_{\mathcal{B}}$ where $\mathcal{B}$ is some basis of V, $n = \dim(\mathcal{B})$, and we call C the coordinate mapping.

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Let $\mathcal{B} = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$, and define $A = \begin{bmatrix} \mathbf{v}_1 & \ldots & \mathbf{v}_n \end{bmatrix}$ to be a matrix whose columns are the basis vectors from $\mathcal{B}$.

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Note: This may not be a matrix of the usual real numbers, could be functions, matrices, etc!

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We will now prove some properties of C!

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$$\mathbf{u} - \mathbf{v} = A\mathbf{x} - A\mathbf{x}$$

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So, $\mathbf{u} = \mathbf{v}$. $\square$

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Let $\mathbf{x} \in \mathbb{R}^n$. See that $A\mathbf{x} = \mathbf{v} \in V$. Then by definition of C, we have that $C(\mathbf{v}) = \mathbf{x}$.

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Let $\mathbf{x} \in \mathbb{R}^n$. See that $A\mathbf{x} = \mathbf{v} \in V$. Then by definition of C, we have that $C(\mathbf{v}) = \mathbf{x}$. Thus, we can reach every element in $\mathbb{R}^n$.

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Let $\mathbf{x} \in \mathbb{R}^n$. See that $A\mathbf{x} = \mathbf{v} \in V$. Then by definition of C, we have that $C(\mathbf{v}) = \mathbf{x}$. Thus, we can reach every element in $\mathbb{R}^n$.
Therefore, C is a surjection $\square$

Coordinate Mapping is a Bijection

Theorem

The coordinate mapping $C : V \to \mathbb{R}^n$ is a bijection.

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The last 2 slides proved that C is an injection and surjection. Therefore it is a bijection. $\square$

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We also call bijections isomorphisms. This just means we can think about the vectors and their coordinate vectors interchangeably!

Coordinate Matrix

Definition

Coordinate Matrix: We define the coordinate matrix of a list of vectors: $\mathbf{v}_1, \ldots, \mathbf{v}_p$ in the basis $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ as follows:

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Example

Consider $V = \mathbb{R}^{2\times 2}, F = \mathbb{R}$ with the basis $\mathcal{B} = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$.

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$$P_{\mathcal{B}} = \begin{bmatrix} 1 & 2 & 0 \\ 4 & 4 & 0 \\ 0 & 1 & 4 \\ 1 & 1 & 2 \end{bmatrix}$$

But Why do we Care?

In weird spaces, it's easier to check for pivot columns in $P_{\mathcal{B}}$ than it is to check if $\mathbf{v}_1, \ldots, \mathbf{v}_p$ are linearly independent!

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Thus, these vectors are linearly independent!